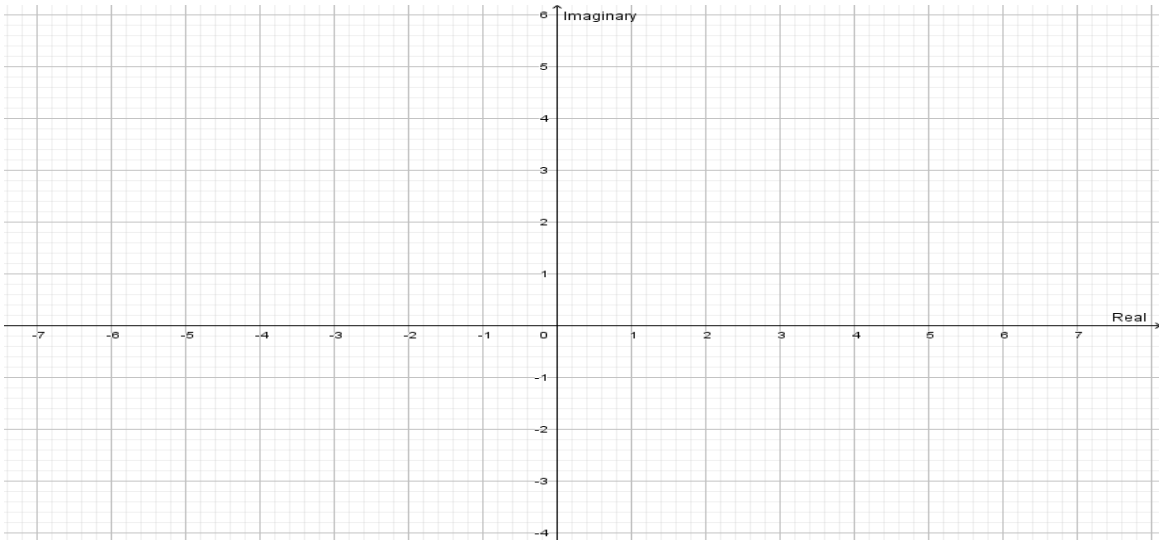
<u>Student Activity</u>: To investigate the complex conjugate



N.B. Note the complex conjugate of a complex number z is written as $\bar{z}$

1. Find the complex conjugate of the following numbers and check your answers using the interactive file.

		Calculate $\bar{z}$.
a.	$z_1 = 3 + 2i$	
b.	$z_1 = 2 + 3i$	
c.	$z_1 = 1 - 3i$	
d.	$z_1 = -1 - 2i$	
e.	$z_1 = -3 + 4i$	
f.	$z_1 = -i$	
g.	$z_1 = 4$	
h.	$z_1 = -1 - i$	
i.	$z_1 = 1 + \sqrt{25}\, i$ and $z_2 = -2 - \sqrt{36}\, i$	

2. On the Argand diagram, which transformation gives the conjugate of a complex number?

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3. If $2-4i$ is the complex conjugate of Z_1, what is Z_1?

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4. Does every complex number have a unique complex conjugate? Explain.

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5. Is the modulus of a complex number equal to the modulus of its complex conjugate? Explain your answer.

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6. What type of number do you always get when you add a complex number to its conjugate?

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7. What can you say about two complex numbers if the sum of these two complex numbers is real?

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8. What type of number do you always get when you subtract a number's complex conjugate from that number?

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9. Multiply $2+3i$ by its complex conjugate. What do you notice?

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10.

   a. Multiply $a + bi$ by its complex conjugate. What do you notice?

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   _____

   b. Is this always the case when you multiply a complex number by its complex conjugate?

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